

Exception Handling

What is an Exception?

* An exception is an event which is occurring during the programming execution and it disturbs the normal flow of the programming execution.

* Because of Exception our programming execution stopped suddenly.

* This sudden termination of the program is called an abnormal termination of the program.

* If any exception is raised in our programming then its handling code is created or not in our programming is checked by our python interpreter.

* If no handling code is available then throws exception otherwise will return user friendly message.

* For handling exceptions, python is providing several keywords. They are try, except, else, finally, as, raise, class keywords.

Interview asking

What is debugging

- + Debugging is the process of finding and fixing errors or bugs in software or hardware
- Programmers use it to identify why a program isn't working as expected and then resolve those issues
- This involves analyzing the code, isolating the problem and making corrections to ensure the final product functions correctly

What is Errors

- An error is a term which is used to describe any issue that arises unexpectedly that cause a computer do not function properly

Run time errors

- The errors which occur at the time of execution of a program are known as runtime errors

- * we get runtime errors because of programming logic, invalid input, memory related issues etc
- * For every runtime error, python is providing corresponding predefined exception class
 - For example `KeyError`, `IndexError`, `ValueError`, `NameError` etc
- At the time of execution of program if any runtime error is occur then internally corresponding `RuntimeError` representation class object will be created
- If python program doesn't contain the code to handle that exception object then our program execution will be terminated abnormally

What is abnormal Termination?

The concept of terminating the program in the middle of its execution without executing the last step of the program is known as abnormal Termination

- It is undesirable situation in any programming language

for example print (10/0) returns ZeroDivision

try, except, else, finally

try

except (1, 2, 3, 4, 5, 6) default

else

finally :

Types of errors

→ Syntax error

→ Runtime error

→ Type error

→ name error

→ key error

→ Index error

→ Value error

→ Attribute error

→ Module not found error

→ Import error

File exist error

File not found error